

Semicon

# “3D ICs Could Be INDIA’S BEST CHANCE To Lead In The Semiconductor Industry”

Cadence Design Systems is collaborating with MeitY to equip universities, schools, and startups with EDA tools. EFY’s Yashasvini Razdan spoke with Jayashankar Narayanankutty about how this alliance strengthens Cadence, along with the company’s strategic growth in India, driven by 3D ICs, RISC-V, and AI-powered chip design.

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## What emerging trends in electronic chip design could encourage more startups to design in India?

The mid-end range of chip technologies offers immense opportunities. For instance, fabs at mature nodes remain highly valuable for volume-based applications. India imports a large number of electronic components and goods, representing opportunities for indigenisation in areas like consumer electronics and automotive components. While high-end chip design offers potential, the mid-end range segment is the sweet spot for value creation and scaling.

To support startups, we may offer tools at a lower upfront cost, with the remainder deferred, making it easier for them to access essential design resources.

## You are focusing on 3D ICs. How do you see India adopting this technology?

If there is one technology where India can immediately take a leadership position, it is 3D ICs. This is because 3D ICs are a relatively new domain globally—no country has extensive domain expertise or established knowledge. If India invests wisely, it could leapfrog and become a leader in this area.

3D ICs are the future, allowing dies manufactured at different nodes (e.g., 2 nanometres and 28 nanometres) to be integrated into a single package. While this reduces costs and enhances performance, it also introduces challenges such as functional verification and routing complexities. Since these challenges are global, India can position itself at the forefront of 3D IC development by addressing them.

I believe 3D ICs could be India’s best chance to lead in the semiconductor industry.

## Are AI-enabled tools accessible to startups and accelerators?

Yes, such tools are accessible to startups, but AI enablement might not initially be a priority for them. At the proof-of-concept stage, where startups invest heavily in developing and validating their ideas, teaching a machine to solve uncharted problems takes time.

AI makes more sense during production, where optimisations and efficiency gains have a more significant impact. While AI is often marketed as a game-changer, its practicality depends on a startup’s specific stage and needs. In the early stages, mastering the basics and focusing on core development is more critical.

## Is AI the only way to reduce time-to-commercialisation for chip designers?

There are other strategies as